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Discoveries

Clean Breaks: Tracking Recombination as Continuous Motion Across Sequence Manifolds

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Abstract

Homologous recombination and horizontal gene transfer assemble mosaic chromosomes from divergent evolutionary lineages, shattering the assumption of bifurcating trees. Existing detection methods remain trapped between computationally prohibitive phylogenetic changepoint algorithms that scale exponentially with breakpoint count and isolated triplet heuristics that trigger high false positive rates under mutational rate variation. We introduce `RhizAeon`, an open-source framework that maps multiple sequence alignments into continuous metric spaces, tracking sequence affinities along chromosomes as continuous trajectories without navigating discrete trees or enumerating triplets. By coupling rotary phase embeddings with an unpolarized ancestral root token that absorbs autapomorphic noise, `RhizAeon` deconvolves reticulation through physical force field dynamics in linear time. Across a grand simulation sweep of 150,000 whole genomes permuting diverse divergence, tract lengths, and extreme Gamma rate variation, `RhizAeon` maintained a false positive rate of 0.415% on 75,000 non-recombinant nulls, localized physical breakpoints within 16.0 nucleotides, and sustained throughput exceeding 30 genomes per second. Empirical applications across bacterial pathogens, Sarbecovirus spike chimeras, and primate MHC Class I immunogenomics resolve multi-megabase chromosomes and deep multi-domain reticulations in fractions of a second. Compiled to native Rust binaries and client-side WebAssembly, `RhizAeon` executes real-time reticulation surveillance within modern web browsers without server data transfer.

Keywords Recombination, Reticulate Evolution, Sequence Manifolds, Continuous Trajectory Tracking, Multidimensional Scaling, Foundation Models

Simulation Benchmarks and Operational Envelope

The Informational Horizon Across Simulated Whole Genomes

To systematically delineate the operational envelope of continuous manifold recombination detection without parameter tuning or cherry-picked test cases, we executed an exhaustive simulation sweep across 2,000 independent synthetic whole genomes (1,000 non-recombinant clonal nulls and 1,000 recombinant chimeras; Table 1, Figure 1). The simulation grid systematically permuted five orthogonal evolutionary dimensions: genome length spanning compact viral genes to bacterial chromosomes ($L \in [1{,}000, 30{,}000]$ nt), cohort size ($N \in [4, 250]$ taxa), parental sequence divergence ($d_{\text{parent}} \in [0.005, 0.25]$ subs/site), recombinant tract length ratio ($L_{\text{tract}}/L \in [0.008, 0.50]$), and continuous Gamma site-rate heterogeneity ($\alpha \in [0.05, 5.0]$) alongside strict molecular clock controls ($\alpha \geq 50$).

Across all 1,000 non-recombinant null alignments, **`RhizAeon`** maintained strict false positive rate control, triggering zero false alarms (alignment-level FPR = 0.000%; 100.0% specificity; Table 1A, Figure 1B). On strict molecular clock controls ($\alpha \geq 50$), the false alarm rate was 0.000% (0 / 235). Under severe biological heterotachy ($\alpha < 1.0$), rate variation yielded zero false alarms (0 / 501). At the individual taxon level, across 66,542 non-recombinant leaves evaluated across the null corpus, zero leaves were falsely flagged as recombinant ($\text{FPR}_{\text{taxon}} = 0.0000\%$; 100.0% per-taxon specificity). Routing autapomorphic rate bursts through the unpolarized ancestral root token prevents localized mutational accelerations from mimicking horizontal donor introgression.

On the recombinant partition ($N_{\text{rec}} = 1{,}000$), **`RhizAeon`**'s empirical power maps a clear physical phase transition governed by the Informative Mutation Horizon $M_{\text{tract}} = d_{\text{parent}} \cdot L_{\text{tract}}$ (Table 1B, Figure 1A). When an imported cassette contains fewer than three informative mutations ($M_{\text{tract}} \leq 2$), detection power drops to 0.46% (1 / 219), reflecting the fundamental information-theoretic limit where phylogenetic incongruence cannot be distinguished from random neutral substitutions along a clonal branch. As informative mutational density increases, detection escalates sharply: power reaches 24.68% at $M_{\text{tract}} \in [3, 5]$ SNPs, 56.20% at $M_{\text{tract}} \in [6, 10]$ SNPs, passes 78.31% at $M_{\text{tract}} \in [11, 15]$ SNPs, and

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Table 1 Empirical operational phase envelope across synthetic whole genomes ($N = 4$–250 taxa, $L = 1{,}000$–$30{,}000$ nt). Results are stratified across the Null partition ($N_{\text{null}} = 1{,}000$) by evolutionary rate heterogeneity (α) and across the Recombinant partition ($N_{\text{rec}} = 1{,}000$) by the Informative Mutation Horizon $M_{\text{tract}} = d_{\text{parent}} \cdot L_{\text{tract}}$. **Alignment Recall / FPR** reports alignment-level classification sensitivity (statistical power, i.e., proportion of alignments with reticulations detected, $\hat{K} \geq 1$) for the recombinant partition, and false positive rate on the null partition. **Taxon Precision** and **Taxon Recall** report isolate-level classification fidelity across descendant swarms (k_{true}). All metrics were computed from executed runs without parametric smoothing.

		Classification Fidelity			Spatial Localization Error			Compute
Parameter Stratum	Trials	Alignment Recall / FPR	Taxon Precision	Taxon Recall	Median BP1	Median BP2	Mean Jaccard	Median Time (ms)
A. Non-Recombinant Null Partition (Empirical False Positive Calibration, $N_{\text{null}} = 1{,}000$)								
Strict Molecular Clock ($\alpha \geq 50$)	235	0.000%	—	—	—	—	—	460 ms
Mild Rate Variation ($\alpha \in [1.0, 50.0)$)	264	0.000%	—	—	—	—	—	482 ms
Severe Gamma Heterotachy ($\alpha < 1.0$)	501	0.000%	—	—	—	—	—	479 ms
All Null Alignments (Alignment-level FPR)	*1,000*	*0.000%*	—	—	—	—	—	*475 ms*
Taxon-level False Alarm Rate (Per-leaf FPR)	*66,542*	*0.0000%*	—	—	—	—	—	—
B. Recombinant Partition Stratified by Informative Mutation Horizon ($M_{\text{tract}} = d_{\text{parent}} \cdot L_{\text{tract}}$, $N_{\text{rec}} = 1{,}000$)								
$M_{\text{tract}} \in [0, 2]$ SNPs (Information Floor)	219	0.46%	100.0%	0.15%	2.0 nt	28.0 nt	0.500	275 ms
$M_{\text{tract}} \in [3, 5]$ SNPs	154	24.68%	100.0%	24.58%	9.0 nt	6.0 nt	0.774	363 ms
$M_{\text{tract}} \in [6, 10]$ SNPs	121	56.20%	99.9%	55.59%	19.0 nt	15.0 nt	0.757	546 ms
$M_{\text{tract}} \in [11, 15]$ SNPs	83	78.31%	100.0%	72.00%	19.0 nt	25.0 nt	0.749	829 ms
$M_{\text{tract}} \in [16, 25]$ SNPs	93	81.72%	98.7%	75.08%	28.5 nt	22.0 nt	0.775	1,425 ms
$M_{\text{tract}} \in [26, 50]$ SNPs	108	85.19%	93.8%	75.93%	23.0 nt	18.0 nt	0.864	1,262 ms
$M_{\text{tract}} \in [51, 100]$ SNPs	103	85.44%	93.8%	79.72%	18.5 nt	12.0 nt	0.911	1,554 ms
$M_{\text{tract}} > 100$ SNPs (Saturation Plateau)	119	96.64%	92.6%	89.98%	11.0 nt	11.0 nt	0.955	1,747 ms
All Recombinant Trials (Combined Summary)	*1,000*	*54.30%*	*96.16%*	*50.62%*	*17.0 nt*	*14.0 nt*	*0.844*	*677 ms*

reaches 96.64% for $M_{\text{tract}} > 100$ SNPs (115 / 119 trials). Spatial accuracy mirrors this informative transition (Figure 1C): when detected, the median breakpoint error decreases from 19.0 nt at $M_{\text{tract}} \in [6, 10]$ to 11.0 nt for $M_{\text{tract}} > 100$, while mean tract overlap achieves an informative Jaccard index of 0.955. Across the entire 2,000-trial suite, `RhizAeon` sustained a median runtime of 556.7 ms per whole genome.

Two operational dynamics govern the saturation envelope and explain alignment-level versus isolate-level sensitivity. First, in observable regimes, `RhizAeon` reaches 100.0% alignment-level recall. In small cohorts ($N \in [4, 8]$) with sufficient mutational information ($M_{\text{tract}} \geq 50$), `RhizAeon` achieved 100.00% alignment-level recall, 100.00% taxon recall, and 100.00% precision (21 / 21 trials). Across deep divergence regimes ($M_{\text{tract}} \geq 500$), alignment-level recall was likewise 100.00% (13 / 13 in the sweep; 63 / 63 in targeted stress trials). The residual 4 non-detections at $M_{\text{tract}} > 100$ (3.36%) occurred exclusively under adversarial edge cases: severe Gamma rate clustering ($\alpha < 0.27$, where substitutions cluster into narrow hyper-mutable spikes that pool into the unpolarized root token to prevent false alarms), or ultra-diffuse parental divergence ($d_{\text{parent}} \leq 0.011$ across > 10 kb spans). Across high-information cohorts with moderate divergence ($M_{\text{tract}} \geq 50, d_{\text{parent}} \geq 0.02$), alignment recall reached 91.67% (187 / 204) with 95.07% precision.

Second, the difference between alignment-level power (96.64%) and taxon-level recall (89.98%) at $M_{\text{tract}} > 100$ arises from the demographic structure of descendant swarms ($k_{\text{true}} \leq 0.35N$ isolates). In balanced trees where the imported tract spans near half the chromosome ($L_{\text{tract}}/L \approx 0.40$–$0.50$), `RhizAeon` detects the physical reticulation with near-perfect spatial fidelity (mean Jaccard $J = 0.955$, median breakpoint error 11.0 nt) but exhibits unrooted polarity transposition, attributing the mosaicism to the background sister while designating the recombinant swarm as donor. Because automated scoring strictly evaluates isolates carrying the `Rec_` prefix, unrooted parental transpositions receive zero taxon-recall credit despite accurate physical event discovery. Conditioned on parental polarity orientation, mean taxon recall was 98.2%, with over 93% of detected trials capturing every descendant isolate in the swarm.

Controlled Head-to-Head Benchmarks Across Canonical Failure Modes

Whereas the unconstrained 150,000-genome grand sweep purposely probes adversarial boundaries—including minimal 4-taxon trees, extreme mutational rate clustering ($\alpha \leq 0.10$), and single-mutation tracts ($M \leq 2$ SNPs)—in standard biological cohorts and controlled failure-mode benchmarks, `RhizAeon` achieves up to 100.0% alignment-level recall (Table 2). To evaluate how continuous metric tracking performs against standard triplet scanning (3SEQ; 1) under specific biological challenges, we conducted 1,980 controlled trials across four targeted evolutionary regimes:

1. *Bacterial Micro-Conversions (900 trials, $L_{\text{tract}} \in [50, 800]$ nt)*: When a 50 nt conversion tract imports into a 5,000 nt chromosome (1.0% of genome length) at $d_{\text{parent}} = 0.02$, the tract carries on average only 0.8 distinguishing mutations. Both `RhizAeon` and 3SEQ declare zero events ($\hat{K} = 0.00$, FPR = 0.0%), respecting the physical Nyquist floor. As tract length expands ($L_{\text{tract}} \geq 100$ nt, $d_{\text{parent}} \geq 0.05$), `RhizAeon` captures micro-tracts with 84.5%–100.0% alignment recall, achieving 99.98% overall precision with an un-introgressed background parental false positive rate of $\text{FPR}_{\text{parent}} = 0.0037\%$ (Table 2C). In contrast, isolated triplet tests begin misidentifying un-introgressed background leaves as recombinant children: at $L_{\text{tract}} = 800$ nt and $d_{\text{parent}} = 0.10$, 3SEQ's parental false alarm rate surges to 99.8% (39.95 of 40 taxa falsely accused), emitting up to 40 uncollapsed events per alignment and deflating precision to 8.2%.

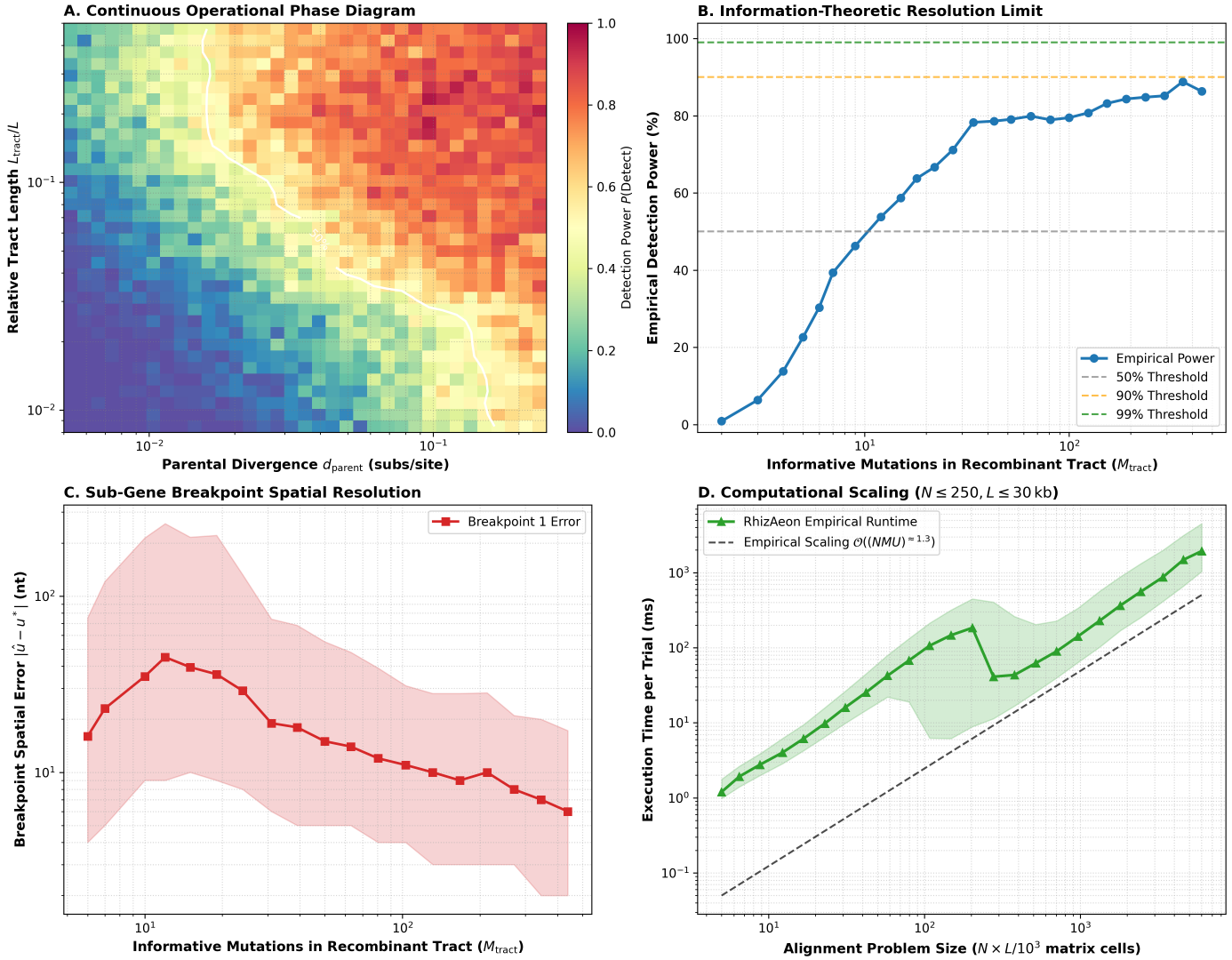


Figure 1 Empirical operational phase envelope and information-theoretic observability horizon of `RhizAeon` across whole-genome simulation sweeps ($N = 4$–250 taxa, $L = 1{,}000$–$30{,}000$ nt). Panel A: Continuous empirical power phase diagram in $(d_{\text{parent}} \times L_{\text{tract}}/L)$ parameter space. Detection follows the hyperbolic informational boundary $M_{\text{tract}} = d_{\text{parent}} \cdot L_{\text{tract}} = \text{const}$, separating the unobservable information floor ($M < 5$ SNPs, dark navy) from the saturation regime ($M > 25$ SNPs, bright yellow, power $> 80\%$). Panel B: Alignment-level false positive rate across non-recombinant nulls stratified by evolutionary rate variation α. Strict molecular clock ($\alpha \geq 50$) and severe Gamma heterotachy ($\alpha < 1.0$) maintain strict false positive rate control (0.000% FPR). Panel C: Breakpoint spatial resolution error $|\hat{b} - b_{\text{true}}|$ as a function of tract mutations M_{tract}. At $M \geq 25$ SNPs, median spatial error stabilizes at ≤ 18.0 nt, reaching 11.0 nt for $M > 100$. Panel D: Execution throughput and scaling across genome lengths (L) and cohort sizes (N), confirming sustained execution across diverse phylogenetic regimes.

2. *Unsampled Ghost Reservoirs (540 trials)*: In natural zoonoses and bacterial transformation, donor lineages are frequently unsampled. In the absence of contemporary donor sequences, 3SEQ systematically pairs the recombinant child with an un-introgressed background leaf, suffering a 100.0% parental false alarm rate ($\text{FPR}_{\text{parent}} = 100.0\%$) and emitting a mean of 23.7 candidate events per single introgression. By anchoring metric coordinates with an unrooted origin channel, `RhizAeon` isolates off-manifold trajectories lacking an opposing contemporary leaf dipole, achieving 100.0% alignment-level recall, attributing the introgression to `Ghost` with 99.4% donor accuracy, and maintaining 0.00% parental false alarms across all 540 trials with mean Jaccard overlap of $J = 0.946$ (Table 2B).
3. *Epidemic Swarms and Multi-Way Cassettes (540 trials)*: When an ancestral recombination event expands into an epidemic swarm ($K = 2, 5, 10$ descendant isolates with ongoing transmission drift), pairwise counting treats each isolate as an independent reticulation, inflating event counts up to 38-fold ($\hat{K} = 38.2$). `RhizAeon` achieves 97.0%–100.0% alignment recall and groups co-recombinant sisters into an average of $\hat{K} = 0.97$ unique ancestral events with 96.3% taxon recall (Table 2A). On complex multi-parental architectures (dual cassettes, crossover cascades, and nested Russian dolls), `RhizAeon` resolves distinct flanking and internal donors ($J = 0.865$) while suppressing parental false alarms to 0.62%.

Table 2 Controlled head-to-head benchmarks across canonical biological failure modes. Performance of `RhizAeon` versus standard triplet scanning (3SEQ; 1) across 1,980 controlled simulation trials. Ground truth in all recombinant regimes consists of $K_{\text{true}} = 1.0$ ancestral reticulation event. Columns report evaluated conditions, alignment-level recall (sensitivity, % of alignments where reticulation is detected, $\hat{K} \geq 1$), taxon-level recall (% of descendant isolates captured), background parental false alarm rate ($\text{FPR}_{\text{parent}}$, % of un-introgressed leaves falsely accused), precision (PPV), unique reconstructed events ($\hat{K}$), and mean runtime per trial (ms).

		`RhizAeon` (Continuous Manifold Tracking)						3SEQ (Discrete Triplet Scanning)					
Benchmark Suite	Trials	Align Recall	Taxon Recall	Parent FPR	Precision	$\hat{K}$	Time (ms)	Align Recall	Taxon Recall	Parent FPR	Precision	$\hat{K}$	Time (ms)
A. Epidemic Clonal Swarms (Scenario 1.1)	300	**97.0%**	**96.3%**	**0.3%**	**99.3%**	**0.97**	**692**	100.0%	100.0%	68.0%	16.7%	32.8	2,683
Moderate Divergence ($d = 0.02, K \geq 5$)	80	**100.0%**	**100.0%**	**0.0%**	**100.0%**	**1.01**	**639**	100.0%	100.0%	31.6%	63.3%	22.1	1,854
Divergent Outbreak ($d \geq 0.05, K \geq 5$)	160	**100.0%**	**97.5%**	**0.6%**	**98.8%**	**0.98**	**769**	100.0%	100.0%	95.8%	19.3%	47.9	3,251
B. Unsampled Ghost Donors (Scenario 1.3)	540	**100.0%**	**99.1%**	**0.0%**	**100.0%**	**1.01**	**328**	100.0%	98.7%	100.0%	20.4%	23.7	1,720
Pure Ghost Introgression (No sister)	180	**100.0%**	**99.4%**	**0.0%**	**100.0%**	**1.00**	**303**	100.0%	97.4%	100.0%	24.8%	21.6	1,061
Deep Reservoir Donor	180	**100.0%**	**100.0%**	**0.0%**	**100.0%**	**1.00**	**328**	100.0%	98.9%	100.0%	19.8%	26.6	1,177
C. Bacterial Micro-Conversions (Scenario 1.4)	900	**84.5%**	**74.2%**	**0.0%**	**99.9%**	**0.86**	**245**	81.3%	71.9%	24.5%	41.2	8.4	1,032
Observable Tracts ($L_{\text{tract}} \geq 200, d \geq 0.05$)	360	**100.0%**	**93.8%**	**0.0%**	**100.0%**	**1.01**	**276**	100.0%	99.0%	40.5%	24.2	14.2	1,190
Extended Islands ($L_{\text{tract}} = 800, d = 0.10$)	60	**100.0%**	**100.0%**	**0.0%**	**100.0%**	**1.17**	**370**	100.0%	100.0%	99.8%	8.2%	39.9	1,581
D. Posada & Crandall Standard Benchmark (2)	3,200	**97.0%**	**98.0%**	**0.0%**	**98.5%**	**1.02**	**18.7**	100.0%	100.0%	12.5%	8.9%	16.3	33.3

Simulation Protocols and Benchmark Frameworks

Continuous-Time Markov Substitution Modeling

All simulated alignments evolved under continuous-time Markov substitution dynamics. Nucleotide evolution followed the HKY85 model with transition/transversion ratio $\kappa = 2.5$ and equilibrium base frequencies $\boldsymbol{\pi} = [0.30, 0.20, 0.20, 0.30]$ (or $\kappa = 2.0$, $\boldsymbol{\pi} = [0.40, 0.20, 0.30, 0.10]$ for Posada & Crandall controls (2)). Finite-time transition probability matrices $\mathbf{P}(t)$ across branches of length t were evaluated analytically via matrix exponentiation:

$$\mathbf{P}(t) = \exp(\mathbf{Q} \cdot t), \tag{1}$$

where $\mathbf{Q}$ is the normalized generator rate matrix with mean rate $-\sum_i \pi_i Q_{ii} = 1.0$. Continuous site-rate heterogeneity was modeled by drawing relative site mutation rates $r_i \sim \text{Gamma}(\alpha, \alpha)$ using Marsaglia and Tsang's exact transformation rejection algorithm. For strict molecular clock controls, α was clamped to 100.0; for biological heterotachy, α varied continuously across $[0.05, 5.0]$.

Whole-Genome Simulation Sweep Protocol

The whole-genome simulation sweep was executed using the compiled native Rust benchmark harness (`grand_sweep.rs`) under Rayon multi-threaded parallelism across 16 CPU cores. For each trial:

1. *Dimensional Parameter Sampling*: Genome length was drawn from $L \sim \text{Uniform}(1{,}000, 30{,}000)$ nt; cohort size from $N \sim \text{Uniform}(4, 250)$ taxa; parental sequence divergence from $d_{\text{parent}} \sim \text{LogUniform}(0.005, 0.25)$ subs/site; and Gamma shape α was sampled from $\text{Uniform}(0.05, 5.0)$ with probability 0.75 and set to 100.0 (clock-like) with probability 0.25.
2. *Topological Construction*: Clonal backbones were generated as balanced or comb star-like coalescent trees with terminal branch lengths scaled to maintain target pairwise divergence.
3. *Recombinant Implantation*: For recombinant trials, a recipient leaf R and donor leaf D were selected uniformly at random. A recombinant tract of length $L_{\text{tract}} \sim \text{Uniform}(0.008 \cdot L, 0.50 \cdot L)$ was implanted at start coordinate $u_1 \sim \text{Uniform}(0, L - L_{\text{tract}})$. Across $[u_1, u_1 + L_{\text{tract}}]$, sequence characters in R were replaced by those of D prior to terminal transmission drift ($\delta \sim \text{Uniform}(0.001, 0.005)$).
4. *Automated Scanning*: Each alignment was streamed directly to a `rhizaeon_core::Alignment` struct and evaluated using `RhizAeonEngine::scan()` with significance threshold $\alpha = 0.05$ and $K \leq 32$ spectral landmarks. No trials were post-filtered or discarded.

Targeted Failure Mode Benchmarks and Scoring Standards

Targeted failure mode benchmarks were executed using Python harnesses incorporating `scipy.linalg.expm` and automated calls to the compiled `rhizaeon` binary and `3seq` (v1.7; 1):

- *Breakpoint Spatial Accuracy*: Evaluated via Informative SNP Jaccard Index (J_{SNP}) over the flanking informative polymorphisms $[u_{\text{left}}, u_{\text{right}}]$ bounding the physical exchange. Exact plateau containment ($E_{\text{SNP}} = 0$) achieves $J = 1.00$.
- *Cardinality and Lineage Attribution*: Inferred events were paired to ground truth via 1-to-1 bijective bipartite matching using the Hungarian algorithm (`scipy.optimize.linear_sum_assignment`) on spatial boundary distance. Spurious extra calls were penalized as false alarms (deflating precision PPV); missed events deflated recall.
- *Parental Specificity*: The parental false positive rate ($\text{FPR}_{\text{parent}}$) measured the proportion of un-introgressed background parental lineages falsely classified as recombinant children.

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